

4.2 & 4.3 Double Integrals Worksheet

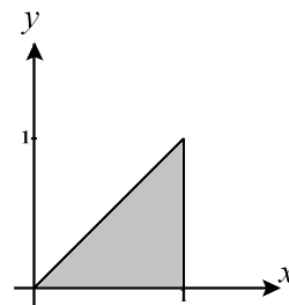
1. Integrate $f(x, y) = x^2 + y^2$ over the domain D , where D is the region bounded by $y = 0$, $x = 1$, & $y = x$.

We sketch a picture of the region D and define D as:

$$D = \{(x, y) \mid 0 \leq y \leq x \text{ \& } 0 \leq x \leq 1\}$$

Therefore, we compute:

$$\begin{aligned} \iint_D f(x, y) dA &= \int_0^1 \int_0^x (x^2 + y^2) dy dx = \int_0^1 \left(x^2 y + \frac{1}{3} y^3 \right) \bigg|_0^x dx \\ &= \int_0^1 \left(x^3 + \frac{1}{3} x^3 \right) dx \\ &= \int_0^1 \frac{4}{3} x^3 dx = \frac{1}{3} x^4 \bigg|_0^1 \\ &= \frac{1}{3} \end{aligned}$$



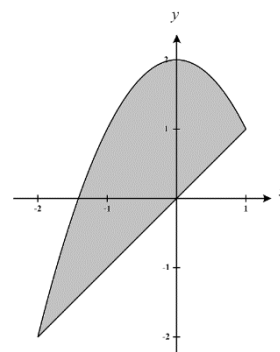
2. Find the volume of the solid that is bounded above by the cylindrical surface $z = x^2$ and below by the domain enclosed by the parabola $y = 2 - x^2$ and the line $y = x$.

We sketch a picture of the region of integration and define D as:

$$D = \{(x, y) \mid x \leq y \leq 2 - x^2 \text{ \& } -2 \leq x \leq 1\}$$

Therefore, we compute:

$$\begin{aligned} V &= \iint_D z dA = \int_{-2}^1 \int_x^{2-x^2} x^2 dy dx = \int_{-2}^1 (x^2 y) \bigg|_x^{2-x^2} dx = \int_{-2}^1 [x^2(2 - x^2) - x^3] dx \\ &= \int_{-2}^1 [2x^2 - x^4 - x^3] dx = \left[\frac{2}{3} x^3 - \frac{1}{5} x^5 - \frac{1}{4} x^4 \right]_{-2}^1 \\ &= \left(\frac{2}{3} - \frac{1}{5} - \frac{1}{4} \right) - \left(-\frac{16}{3} + \frac{32}{5} - \frac{16}{4} \right) = \frac{63}{20} \end{aligned}$$



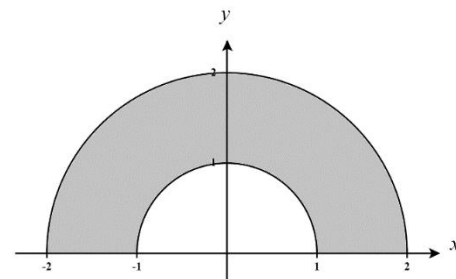
3. Use a double integral to determine the area of the region given by $x^2 + y^2 \geq 1$ & $x^2 + y^2 \leq 4$ which exists above the x -axis.

We sketch a picture of the region of integration and define D in polar coordinates:

$$D = \{(r, \theta) \mid 1 \leq r \leq 2 \text{ \& } 0 \leq \theta \leq \pi\}$$

Therefore, we compute:

$$\begin{aligned} A &= \iint_D 1 dA = \int_0^\pi \int_1^2 1 r dr d\theta = \int_0^\pi \left[\frac{1}{2} r^2 \right]_1^2 d\theta = \int_0^\pi \left[2 - \frac{1}{2} \right] d\theta \\ &= \left[\frac{3}{2} \theta \right]_0^\pi \\ &= \frac{3\pi}{2} \end{aligned}$$



4. A cylindrical drill bit is used to bore a vertical hole through the center of the solid bounded above by the graph of $f(x, y) = 25e^{-(x^2+y^2)/4}$ and bounded below by the region in the xy -plane given by $x^2 + y^2 \leq 16$. What is the radius of the drill bit if 10% of the volume of the solid is removed?

If we can determine the volume of the entire solid then we can easily determine the remaining volume after 10% has been removed. To find the volume of the entire solid we begin by sketching a picture of the region of integration and D in polar coordinates.

We define D in polar coordinates: $D = \{(r, \theta) \mid 0 \leq r \leq 4 \text{ \& } 0 \leq \theta \leq 2\pi\}$

Therefore, the volume of the entire solid is given by:

$$V = \iint_D f(x, y) dA = \int_0^{2\pi} \int_0^4 25e^{-(r^2)/4} r dr d\theta$$

We proceed by doing a u -sub. Let $u = -\frac{r^2}{4}$ then $du = -\frac{1}{2} r dr \rightarrow -2du = r dr$.

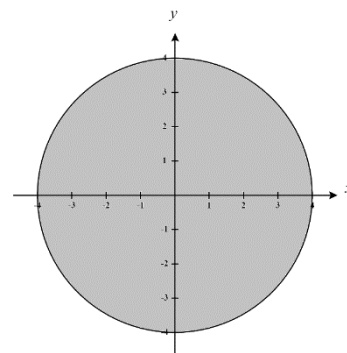
Our bounds become $u(0) = 0$ & $u(4) = -4$. So we get,

$$\int_0^{2\pi} \int_0^4 25e^{-(r^2)/4} r dr d\theta = \int_0^{2\pi} \int_0^{-4} -50e^u du d\theta = \int_0^{2\pi} -50e^u \Big|_0^{-4} d\theta = \int_0^{2\pi} (-50e^{-4} + 50) d\theta = 2\pi(50 - 50e^{-4})$$

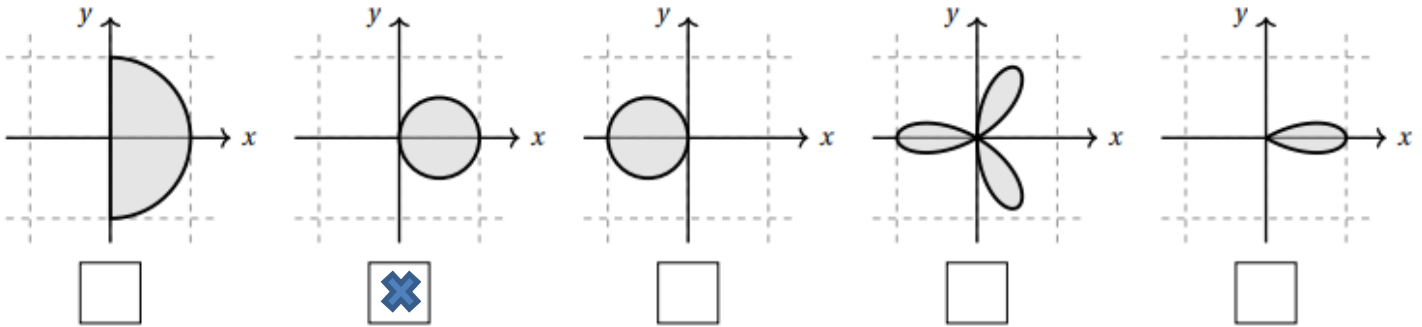
Therefore the volume of the drilled out portion needs to be $\frac{1}{10} [2\pi(50 - 50e^{-4})] = 10\pi(1 - e^{-4})$.

The cylinder that was drilled out was located at the origin and at this point the surface has a height of $f(0, 0) = 25e^0 = 25$ and therefore our cylinder here has a height of 25. This means that the volume equation for this cylinder can be written as $10\pi(1 - e^{-4}) = \pi r^2(25)$. This implies that

$$r^2 = \frac{2}{5}(1 - e^{-4}) \rightarrow r = \sqrt{\frac{2}{5}(1 - e^{-4})}$$



5. The integral $\iint_R (2x^2 + 2y^2 + y) dA$ has the form $\int_{-\pi/2}^{\pi/2} \int_0^{\cos(\theta)} ?? dr d\theta$ when converted into polar coordinates.
- a. Mark the box below with the picture of the region that represents R .



- b. Fill in the missing integrand to convert this integral into polar coordinates.

$$\iint_R (2x^2 + 2y^2 + y) dA = \int_{-\pi/2}^{\pi/2} \int_0^{\cos(\theta)} (2r^2 + r \sin(\theta)) r dr d\theta$$

6. Determine the volume of the region that lies under the sphere $x^2 + y^2 + z^2 = 9$, above the plane $z = 0$ and inside the cylinder $x^2 + y^2 = 5$.

We begin by sketching a picture of the solid in question here.

To find the volume of the entire solid we see that we are working under the sphere $x^2 + y^2 + z^2 = 9$ and above the disk bounded by $x^2 + y^2 = 5$ in the xy -plane. Since we are integrating over the disk $x^2 + y^2 = 5$ we will approach this question using polar coordinates. Let D represent the disk bounded by $x^2 + y^2 = 5$ and note that $z = \sqrt{9 - x^2 - y^2}$

We define D in polar coordinates:

$$D = \{(r, \theta) | 0 \leq r \leq \sqrt{5} \text{ \& \; } 0 \leq \theta \leq 2\pi\}$$

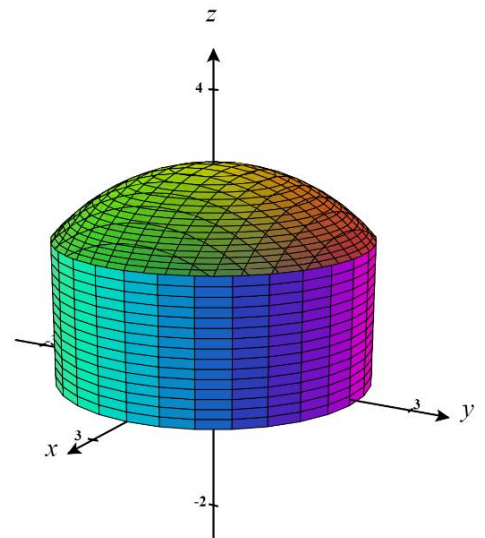
Therefore, the volume of the entire solid is given by:

$$V = \iint_D f(x, y) dA = \int_0^{2\pi} \int_0^{\sqrt{5}} \sqrt{9 - r^2} r dr d\theta$$

We proceed by doing a u -sub. Let $u = 9 - r^2$ then $du = -2r dr \rightarrow -\frac{1}{2} du = r dr$.

Our bounds become $u(0) = 9$ & $u(\sqrt{5}) = 4$. So we get,

$$\int_0^{2\pi} \int_0^{\sqrt{5}} \sqrt{9 - r^2} r dr d\theta = \int_0^{2\pi} \int_9^4 -\frac{1}{2} \sqrt{u} du d\theta = \int_0^{2\pi} \left[-\frac{1}{3} u^{3/2} \right]_9^4 d\theta = \int_0^{2\pi} \left(-\frac{1}{3} (8) + \frac{1}{3} (27) \right) d\theta = 2\pi \left(\frac{-8}{3} + 9 \right) = \frac{38\pi}{3}$$



7. Use a double integral to determine the area of the region bounded by the curves $y = \sqrt{x}$ and $y = x^2 + 1$ on the interval from $[1, 2]$.

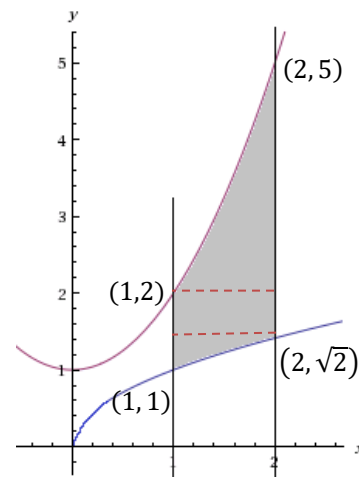
We sketch a picture of the region of integration D and define D as:

$$D = \{(x, y) \mid \sqrt{x} \leq y \leq x^2 + 1 \text{ \& } 1 \leq x \leq 2\}$$

(Note how difficult it would be to define this region if we wanted to setup differentials in the order $dx dy$.)

Therefore, we compute:

$$\begin{aligned} A &= \iint_D 1 dA = \int_1^2 \int_{\sqrt{x}}^{x^2+1} 1 dy dx = \int_1^2 (y) \Big|_{\sqrt{x}}^{x^2+1} dx \\ &= \int_1^2 (x^2 + 1 - \sqrt{x}) dx \\ &= \left(\frac{1}{3} x^3 + x - \frac{2}{3} x^{3/2} \right) \Big|_1^2 \\ &= \left(\frac{8}{3} + 2 - \frac{2}{3} \sqrt{8} \right) - \left(\frac{1}{3} + 1 - \frac{2}{3} \right) \\ &= 4 - \frac{2}{3} \sqrt{8} \end{aligned}$$



8. Evaluate the integral $\int_{-1}^1 \int_{-\sqrt{1-x^2}}^0 \cos(x^2 + y^2) dy dx$.

Trying to tackle this question in cartesian coordinates will be difficult, therefore we will attempt to convert to polar coordinates. We see that the region of integration D can be defined as:

$$D = \{(x, y) \mid -1 \leq x \leq 1 \text{ \& } -\sqrt{1-x^2} \leq y \leq 0\}$$

We recognize the lower bound on y as the bottom half of a circle of radius 1 centered at the origin and therefore the region D looks like the region shown at the right.

We then redefined D using polar coordinates:

$$D = \{(r, \theta) \mid 0 \leq r \leq 1 \text{ \& } \pi \leq \theta \leq 2\pi\}$$

Now, we compute the integral as follows:

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^0 \cos(x^2 + y^2) dy dx = \int_{\pi}^{2\pi} \int_0^1 \cos(r^2) r dr d\theta = \int_{\pi}^{2\pi} \frac{1}{2} \sin(r^2) \Big|_0^1 d\theta = \int_{\pi}^{2\pi} \frac{1}{2} \sin(1) d\theta = \frac{\pi}{2} \sin(1)$$

